Bittle X Platform v1.3 - Interface Specification

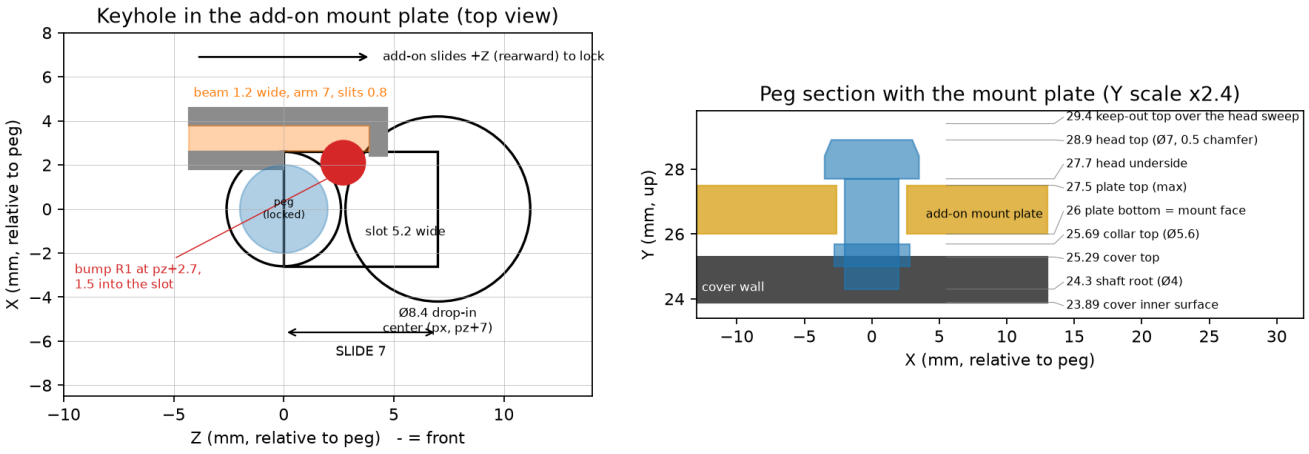
Version 1.3, 2026-10-15. The platform is the stock Petoi Bittle X back cover with 4 pegs on its top. An add-on has a plate with 4 keyholes. Put it on the pegs, slide it 7 mm rearward, and it clicks. No tools, glue or tape are necessary. interface.json has all the numbers in machine-readable form.



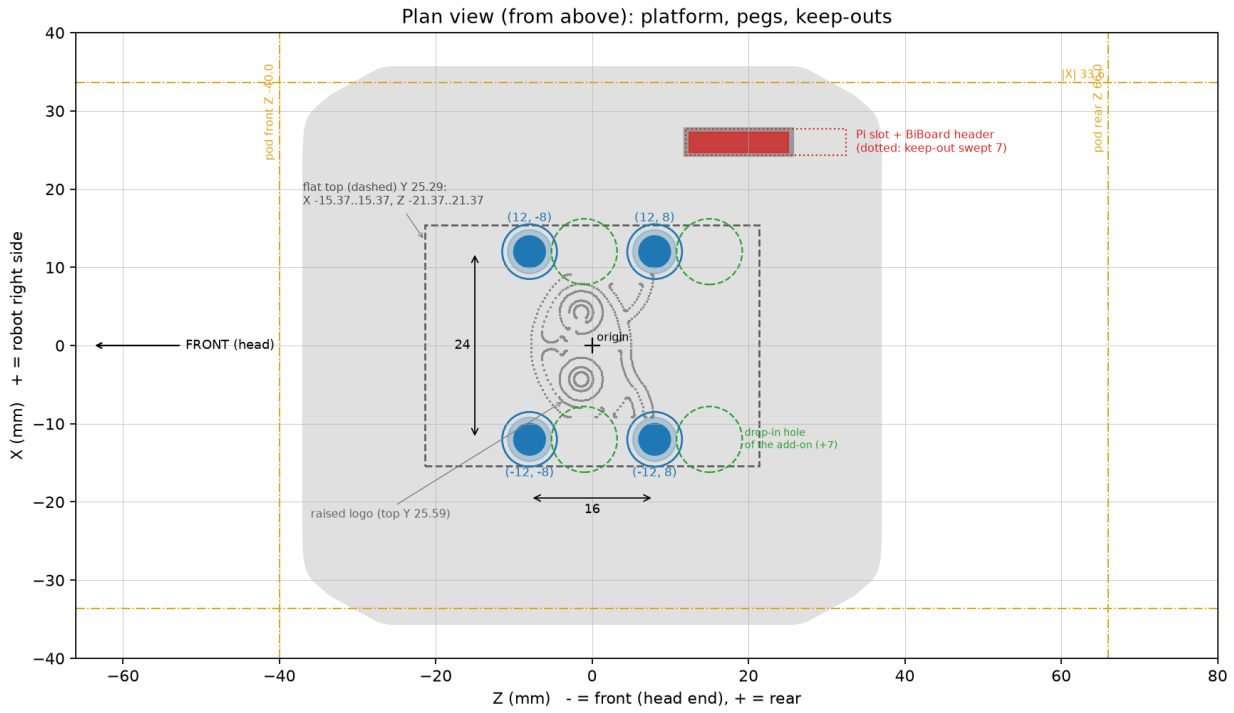
The platform (dark) with the starter plate (yellow).

1. Key numbers

Item	Value (mm)
Frame	X lateral (+X = robot right), Y up, Z along the body (-Z = head end). Origin: center line, middle of the cover.
Pegs (X, Z)	(12, -8), (12, 8), (-12, -8), (-12, 8): a centered 24 x 16 rectangle
Peg	shaft Ø4; head Ø7, Y 27.7 to 28.9
Add-on plate	bottom at Y 26, 1.5 thick at the keyholes
Keyhole	slot 5.2 wide from the peg to 7 mm behind it (+Z); Ø8.4 drop-in hole at the rear end
Fit	put the add-on on 7 mm forward of its place, then slide it 7 mm rearward until it clicks
Click (detent)	a 1.2 wide spring beam on the +X side of each slot, freed by a 0.8 U slit (fixed at its front end only); bump R1, 1.5 into the slot, 2.7 behind the peg
Keep clear	the peg heads as they slide; the BiBoard header in the cover slot (X 24.73..27.27, Z 12.4..25.1, top Y 29.1)
Size	the pod fits in Z -40..66, |X| <= 33.6; outside that, check the legs and the head



Left: keyhole and click beam in the add-on plate (top view). Right: the peg and the plate (side section).



Plan view: pegs (blue), drop-in holes (green), BiBoard header (red), pod size (gold).

2. Design an add-on

1 Start from cad/starter_plate.step. It has the keyholes and the click beams.

2 Add your part on top of the plate, in the same print. You can also screw parts to the M3 holes.

3 Keep your part out of stl/installed/keepout.stl. This keeps it clear of the robot, the peg heads and the header.

4 Keep heavy parts low and near the pegs. Keep the center of mass of the add-on within 10 mm of Z 0, near X 0.

5 Keep the weight low and near the center line. The pegs carry the whole add-on.

6 For a plate of a different shape: subtract cad/keyhole_cutter.step, then add cad/detent_bumps.step. Both are in position.

3. Printing

Part	Orientation	Supports	Cura profile
cover_pegs_foot (recommended)	stand on the rear end, foot on the bed	none: the foot is the support	Bittle Cover - PLA 0.25, brim off. Snap off the foot after the print.
cover_pegs (no foot)	stand on the rear end	tree, from the bed	Bittle Cover - PLA 0.25 (14-line brim)
starter_plate and add-ons	plate bottom on the bed	none	Bittle Pi Pod - PLA 0.25
small, detailed parts	-	-	Bittle Fine - PLA 0.10

Setting	0.25 profiles	Fine 0.10
Layer	0.25	0.10 (first layer 0.25)
Walls / infill	3 / 20 % gyroid	3 / 20 % gyroid
Outer wall / inner wall / infill speed	30 / 40 / 60 mm/s	25 / 35 / 50 mm/s
Outer wall acceleration	600 mm/s²	500 mm/s²
Nozzle	220 C (PLA)	220 C (PLA)

1 Material: PLA works. Use PETG for add-ons that you remove often or that get warm.

2 Standing on its end, the cover prints the pegs with layers along their length. This makes them strong.

3 The cover stands on a thin strip, and its center of mass is past that strip. The foot is a 3-layer plate with 7 fins under the leaning face. It increases the first layer from 53 to about 1000 mm². The fins touch the cover only at small teeth.

4 In add-ons, keep overhangs at 45 deg or less. Use chamfers, not round edges, on the bottom side.

5 The fits use 0.2 to 0.3 clearance. Calibrate your printer if the plate does not slide on.

4. Files

File	Content
stl/print/	the parts to print, in print orientation
cad/starter_plate.step	the start point for an add-on
cad/pegs.step, keyhole_cutter.step, detent_bumps.step	the interface parts, in position (robot frame)
stl/installed/	all parts and keepout.stl in the robot frame
interface.json / interface.py	all numbers
build.py, docs.py	rebuild the parts and this document (build123d, manifold3d)

Open items: the BiBoard header height is an estimate from photos. The legs, the head and the robot balance are not modeled. The Petoi cover is a mesh, so cover_pegs is STL only.

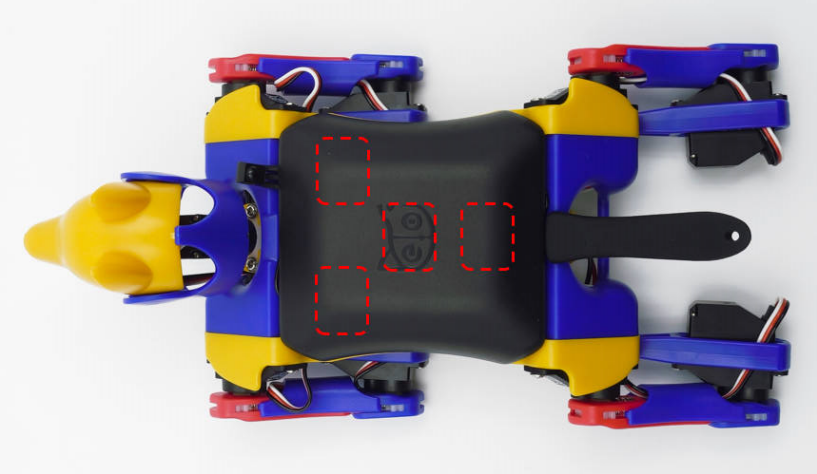
5. Bittle X V2 back touch sensor

1 On a Bittle X V2, the back touch sensor is a flexible circuit that is glued to the inside of the cover top. It has touch areas on the left, the right, the center and near the tail (docs.petoi.com/extensible-modules/back-touch-sensor).

2 The peg roots stop inside the 1.4 mm wall, so the inside of the cover does not change. Move the circuit from the stock cover to cover_pegs in the same way as on the stock cover.

3 An add-on on the platform covers most of the touch areas. Disable the back touch mode (serial command Xb) when an add-on is on the robot.

4 For this reason, the platform has no parts under the cover top. A cover that prints flat without supports needs pegs held from inside the cover, where the circuit is. A possible interface v2 with sockets in a thicker top avoids this.



Bittle X V2 touch areas (red boxes). Image: Petoi, docs.petoi.com/extensible-modules/back-touch-sensor.

6. License

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